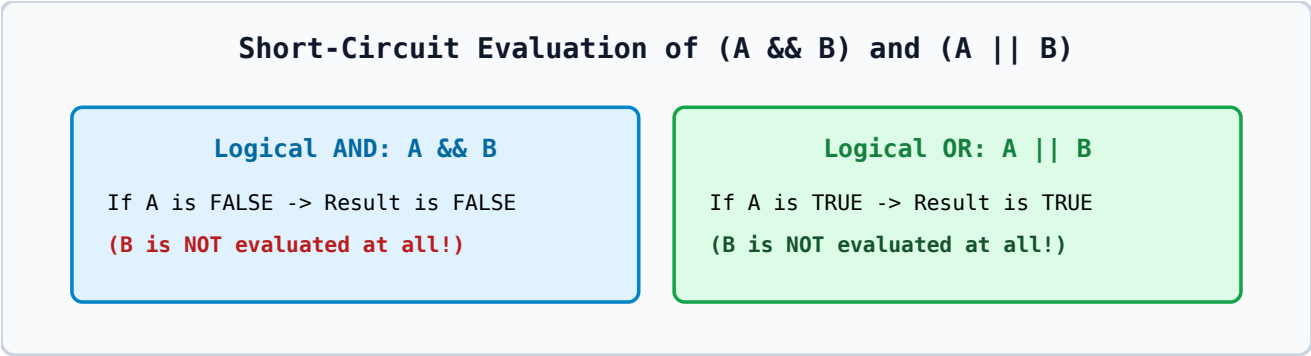
03 OPERATORS AND EXPRESSIONS • THEORY

01 arithmetic relational and short circuit

Detailed Theory, Exact C++ Rules, Memory Models & Exam Reference

01 — Arithmetic, Relational, and Short-Circuit Logical Operators



Short-Circuit Evaluation

1. Integer Division vs. Floating-Point Division

- When **both** operands of `/` are integers (`7 / 2`), C++ performs **integer division**, truncating toward zero (`3`).
- The remainder operator `%` works **only** on integer types (`7 % 2 == 1`).

2. Short-Circuit Evaluation of `&&` and `||`

Unlike `+` or `*`, the built-in logical `&&` and `||` operators guarantee strict **Left-to-Right** evaluation with **short-circuiting**:

- In `expr1 && expr2`: If `expr1` evaluates to `false`, `expr2` is **not evaluated at all**! (This makes `if (p != nullptr && *p == 10)` completely safe!).
- In `expr1 || expr2`: If `expr1` evaluates to `true`, `expr2` is **not evaluated at all**!